

Functions of Several Variables

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Let us begin by visiting some standard definitions.

A point in $\mathbb{R}^3$ is determined by an ordered triple (a, b, c) .

The points $(1, 1, 1)$, $(1, -2, 3)$ and $(-1, 2, -5)$.

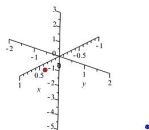


Figure: The points $(1, 1, 1)$, $(1, -2, 3)$ and $(-1, 2, -5)$.

- We use the "right hand rule" to construct the z-axis
- There are 3-coordinate planes: xy-plane, yz-plane and xz-plane
- There are 8 octants.
 - In the first octant, we have: $x, y, z \geq 0$.

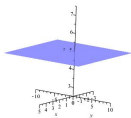
We can project a point $P(a, b, c)$ onto the three coordinate planes.

Definition. We define

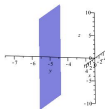
$$\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R} := \{(x, y, z) : x, y, z \in \mathbb{R}\}.$$

Example Sketch the following surfaces in $\mathbb{R}^3$.

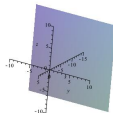
- $z = 5$, $y = -5$ and $x = -10$.



(a) $z = 5$



(b) $y = -5$

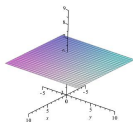


(c) $x = -10$

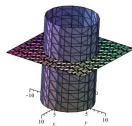
Example Sketch the following surfaces in $\mathbb{R}^3$.
 $x^2 + y^2 = 36$ and $z = 6$.



(a)
 $x^2 + y^2 = 36$



(b) $z = 6$



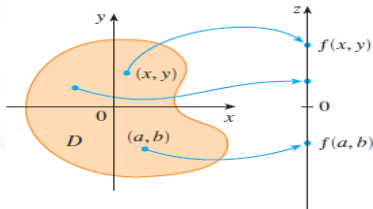
(c)
 $x^2 + y^2 = 36$
and $z = 6$

Note that $x^2 + y^2 = 36$ is a circle in $\mathbb{R}^2$ but a cylinder in $\mathbb{R}^3$.

A function of two variables ($z = f(x, y)$) is a rule that assigns each ordered pair of real numbers (x, y) in a set $D \subset \mathbb{R}^2$ to a unique real number $f(x, y) \in \mathbb{R}$. In this setting:

D is called the **domain** of the function. The set of all possible values of $f(x, y)$ is called the **range** of the function.

For the function $z = f(x, y)$, x and y are called the **independent variables** and z is called the **dependent variable**.



Similarly we can define functions with three variables
 $w = f(x, y, z)$ or even functions with n -variables
 $z = f(x_1, x_2, \dots, x_n)$.

If $z = f(x, y)$, then we define **the graph of f** by the set

$$G_f := \{(x, y, f(x, y)) \in \mathbb{R}^3 : (x, y) \in \text{Dom}(f)\}$$

Note that the graph of

$y = f(x)$ is in $\mathbb{R}^2$, two-dimensional space

$z = f(x, y)$ is in $\mathbb{R}^3$, three-dimensional space

$z = f(x_1, x_2, \dots, x_n)$ in $\mathbb{R}^{n+1}$, in the $(n+1)$ -dimensional space

Example 1. Consider the function

$$z = f(x, y) = \sin(2x + 5y).$$

- Find and sketch the domain of f
- Find the range of f

Solution.

- $\text{Dom}(f) = \mathbb{R}^2$
- Since the sine function takes values between -1 and 1 , we have $\text{Ran}(f) = [-1, 1]$.

Example 2. Consider the function

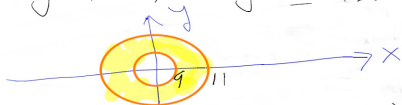
$$f(x, y) = x^2 - \tan^{-1}(4y) + \sqrt{x^2 + y^2 - 81} - 2\sqrt{121 - x^2 - y^2}.$$

- Find and sketch the domain of f

Solution.

$$x^2 + y^2 - 81 \geq 0 \Rightarrow x^2 + y^2 \leq 81$$

$$121 - x^2 - y^2 \geq 0 \Rightarrow x^2 + y^2 \leq 121$$



(The region between two circles).

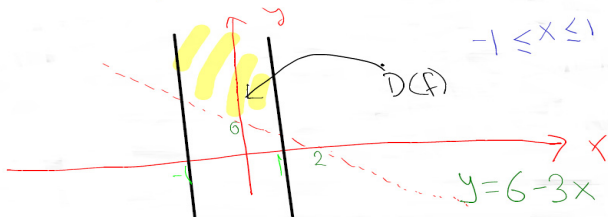
Example 3. Consider the function

$$f(x, y) = \ln(6x + 2y - 12) - \sqrt{1 - x^2}.$$

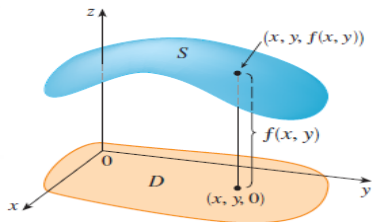
- Find and sketch the domain of f

Solution.

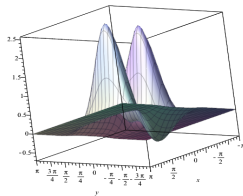
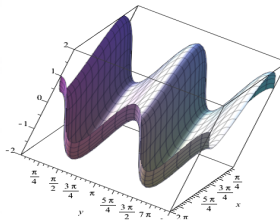
$$6x + 2y - 12 > 0 \Rightarrow 2y > 12 - 6x \Rightarrow y > 6 - 3x$$
$$1 - x^2 \geq 0 \Rightarrow x^2 \leq 1 \Rightarrow \sqrt{x^2} \leq \sqrt{1} \Rightarrow |x| \leq 1 \Rightarrow$$



Sketching the graph of a function $z = f(x, y)$ can be a tedious job.



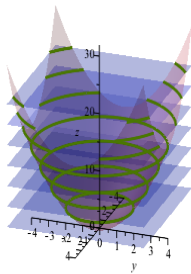
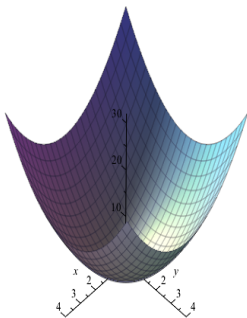
For example, the graphs of $z = \sin(x) + \cos(2y)$ and $z = (x^3 + 5y^2)2^{-x^2-y^4}$ are shown below.



In order to sketch the graph of $z = f(x, y)$ we make use of **level curves**.

Definition The level curves of a function $z = f(x, y)$ are the curves with equations $f(x, y) = k$, where k is a constant ($k \in \text{Range}(f)$). A family of level curves obtained by different values of k is called a **contour map** of f .

Example 4. The graph of $z = x^2 + y^2$ and its level curves are as follows.



Example 5. Consider $z = f(x, y) = \sqrt{25 - x^2 - y^2}$. Find the domain and the range of f . Draw a contour map of f by sketching the level curves of the function for the values $k = 0, 1, 2, 3, 5$. Finally, sketch the graph of f .

Solution.

$$25 - x^2 - y^2 \geq 0 \Rightarrow x^2 + y^2 \leq 25$$

$$R(f) = [0, 5]$$

$$z = k \Rightarrow \sqrt{25 - x^2 - y^2} = k \Rightarrow 25 - x^2 - y^2 = k^2 \Rightarrow$$

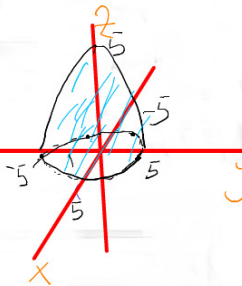
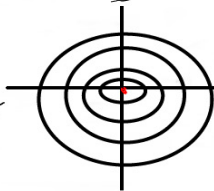
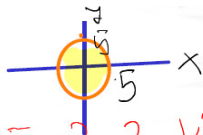
$$x^2 + y^2 = 25 - k^2, \text{ circles}$$

$$k = 0 \Rightarrow x^2 + y^2 = 25$$

$$k = 1 \Rightarrow x^2 + y^2 = 24$$

!

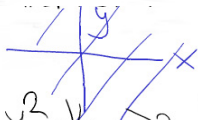
$$k = 5 \Rightarrow x^2 + y^2 = 0 \Rightarrow x = y = 0$$



Example 6. Consider $z = f(x, y) = 7 + 3x^2 + 5y^2$. Find the domain and the range of f . Draw a contour map of f by sketching several level curves of the function. Finally, sketch the graph of f .

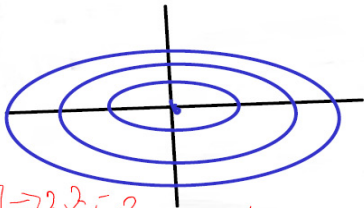
Solution.

$$D(f) = \mathbb{R}^2$$



$$R(f) = [7, \infty)$$

$$z = k \Rightarrow 7 + 3x^2 + 5y^2 = k \Rightarrow 3x^2 + 5y^2 = k - 7, \quad k > 7$$



$$k = 7 \Rightarrow 3x^2 + 5y^2 = 0 \Rightarrow x = y = 0$$

$$k = 8 \Rightarrow 3x^2 + 5y^2 = 1$$



Note that in case of a function of 3 variables
 $w = f(x, y, z)$:

$$f(x, y, z) = k \quad (\text{where } k \in \text{Ran}(f))$$

we obtain **level surfaces**.

Example 7. Find the domain and range of $w = f(x, y, z) = x^2 + y^2 + z^2$.
What are the level surfaces of it?

Solution.

